IBCO-Net: Integrity-Boundary-Corner Optimization in a General Multistage Network for Building Fine Segmentation From Remote Sensing Images

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***Abstract*— Building extraction is a significant topic in high-resolution remote sensing. Insufficient integrity, irregular boundaries, and inaccurate corners remain a problem for existing methods. However, individually optimizing one of these aspects may leave problems in others. Unfortunately, few methods consider integrity, boundary, and corner simultaneously. In this study, we propose a three-stage network [integrity-boundary-corner optimization in a general multistage network (IBCO-Net)] incorporating integrity-boundary-corner optimization for fine segmentation of buildings. First, long-range dependent and spatial-continuous (LDSC) blocks are plugged into the decoder to enhance building integrity. Second, the direction field correction module (DFCM) controls the overall shape of the building by learning the direction field and executing an iterative correction algorithm. Finally, the multistrategy point refinement module (MSPRM) selects boundary and corner points for reclassification to further refine the boundary and relocate corners, and a hybrid loss function supervises IBCO-Net to optimize each stage. Comparative experiments were conducted on three datasets: the Massachusetts building dataset, the ISPRS Potsdam dataset, and the dataset of building instances of typical cities in China. We evaluated common pixel-level metrics and object-level boundary and corner metrics, with experimental results showing that IBCO-Net outperforms eight state-of-the-art convolution neural network (CNN) and transformer-based methods. In addition, the generality of the proposed method is demonstrated via its performance by applying nine existing backbone networks.**

***Index Terms*— Boundary and corner optimization, building extraction, high-resolution remote sensing, multistage network, semantic segmentation.**

I. Introduction

With accelerated urbanization and land development, it has become significant to extract building information from remote sensing images for policy making, population estimation, and smart city construction [1], [2], [3]. Manual labeling of buildings from images is time-consuming, labor-intensive, and tedious. The increased resolution of remote sensing images can enable classification at the building scale [4]. Various automated and semiautomatic methods [5], [6], [7], [8] have been proposed for the rapid and accurate extraction of buildings. Among these methods, semantic segmentation-based methods that classify each pixel have gradually become mainstream due to their excellent performance.

In recent years, the availability and accessibility of remote sensing images have been improved. Deep learning techniques driven by abundant data, such as convolution neural networks (CNNs) [9] and fully convolution networks (FCNs) [10], are increasingly being used for semantic segmentation [11], object detection [12], and scene classification [13] on remote sensing images. In particular, due to the tremendous success of the FCN in the semantic segmentation task, it is widely used in the segmentation of building footprints. Moreover, researchers have proposed methods to optimize building segmentation. For example, some methods enhance multiscale building extraction by multilevel feature fusion [14], [15], [16], [17] and receptive pyramids [18], [19]. Some methods refine the boundaries by edge detection [18], enhancement [19], or postprocessing techniques [16], [20], [21].

However, the existing building segmentation methods could be strengthened to exploit the building's characteristics, as they generally optimize only one building element to neglect others. Therefore, there are still some challenging problems, which we summarize as follows.

1) The diversity of shapes, sizes, materials, and textures makes it difficult for models to recognize each building as a complete individual. Especially, models cannot completely perceive some large buildings that dominate the image, yielding some segmentation results with internal holes, as shown in Fig. 1(c).
2) Spatial convolution and pooling can cause irregular boundaries and inaccurate corners due to the loss of precise location information, while minor errors or missing detections along the boundaries can worsen the situation. Although some methods are designed to improve the building's integrity, they still have defects in boundaries and accurate corners, as shown in Fig. 1(d).
3) Existing boundary optimization methods improve on irregular and inaccurate boundaries. Still, they only deal with boundary pixels occupying a small portion of the building, ignoring the connection between the main body

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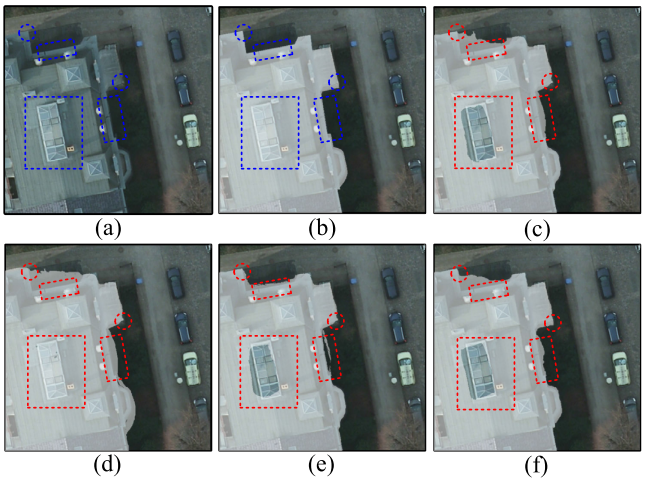


Fig. 1. Typical defects of existing building segmentation methods. (a) Original image. (b) Ground truth. (c) Buildings with holes. (d) Integrity's individual optimization alleviates (c) but leaves irregular boundaries and inaccurate corners. (e) Individual optimization of the boundaries deals with a small portion of pixels and is of limited use for integrity. (f) Individual optimization of corners ignores integrity and boundaries.

and the boundary. This drawback leaves holes inside the building and tends to produce broken and isolated boundaries [see Fig. 1(e)].

4) The building outline can be simplified into a sequence of corner vertices, so the corners are essential for representing the building morphology. However, corner optimization has not received as much attention from the existing methods as integrity and boundary. Individually optimizing the integrity or boundary tends to ignore and leave inaccurate corners, as in Fig. 1(d) and (e). Similarly, individual optimization of corners also neglects the integrity and boundary, as in Fig. 1(f).

Motivated by the above issues, this study is devoted to establish a paradigm applicable to the existing models to incorporate building integrity, boundary, and corner optimization. The corresponding method is a three-stage network under hybrid loss supervision. The three stages of the proposed method are successively carried out for the building integrity optimization, the overall shape control for boundary optimization, and the refinement of the boundary and corner points. The contribution of this work is summarized as follows.

1) We propose a flexible integrity-boundary-corner optimization paradigm for segmenting buildings from remote sensing images. It is applicable to a variety of existing advanced CNN and transformer-based backbone networks. Unlike a simple application of deep learning, the proposed integrity-boundary-corner optimization in a general multistage network (IBCO-Net) is a deep learning-artificial algorithm-integrated method. The results show that the proposed method performs better than the existing state-of-the-art methods.
2) Integrity is an important part of building extraction performance and is the basis for boundary and corner optimization. Plug-and-play long-range dependent and spatial-continuous (LDSC) blocks are proposed to be embedded into the decoder. The cascading sparse long-range dependence and enhanced spatial continuity improve building integrity while recovering detailed information.
3) Controlling the overall shape to optimize the boundary. The direction field correction module (DFCM) is designed to predict the direction field and utilizes an iterative algorithm to correct errors in the feature map. Learning the defined body-boundary linked direction field implicitly facilitates the connection between the building body and the boundary.
4) Boundary and corner points are further refined. We designed the multistrategy point refinement module (MSPRM) that can select points using multiple rational rules and reclassify them. MSPRM incorporates the artificial algorithm's ability to efficiently select points and the deep learning technology's powerful classification capability. It can not only further refine the boundaries but also relocate corners to maintain the building's real morphology.

II. Related Work

A. Integrity Optimization of Building Segmentation

For semantic segmentation methods, the extraction of multiscale features and awareness of multiscale information are the main means to improve the integrity of buildings. For example, Sun et al. [22] designed a multiscale CNN using convolutional kernels of different sizes to extract features in parallel. Xie et al. [23] derive predictions for each feature map in the decoding stage and obtain multiple losses. However, there is a lack of correlation between the features or predictions obtained in parallel, and it is difficult to establish a reasonable supervision strategy. Besides, some pyramidal structures have also been proposed for multiscale context awareness, such as spatial pyramid pooling (SPP) [24] and atrous SPP (ASPP) [16]. Because of their effectiveness, a similar structure has been introduced into networks for building extraction [25]. These pyramidal structures are typically deployed as bridges between the encoder and decoder to perceive multiscale contextual information. However, since they contain pooling operations that tend to lose local information, this results in blurred boundaries and missing corners.

B. Boundary Optimization of Building Segmentation

In both generic semantic segmentation and building segmentation, the boundary quality is a critical aspect reflecting the performance. Techniques, such as edge-guided semantic segmentation [26], point rendering [27], high-low-level feature aggregation [28], high-frequency awareness [29], and subpixel mapping [30], have been shown to improve segmentation and classification quality. In recent years, more and more researchers have paid attention to building boundary optimization. They work to alleviate the blurred and irregular boundaries to achieve a more realistic representation of the building morphology. For instance, Yang et al. [31] combine building footprint extraction with edge perception networks supervised by an edge-aware loss function. Jung et al. [18] enhance the extraction and representation of boundary features

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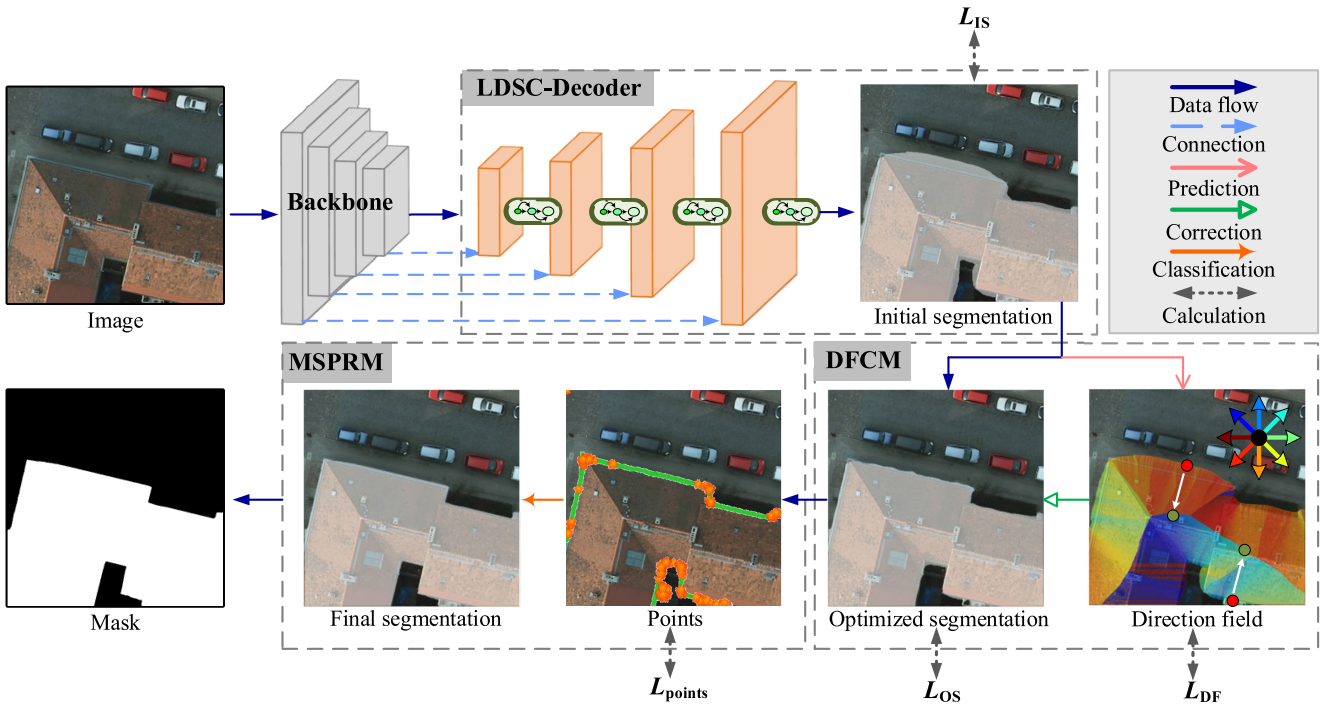


Fig. 2. Overview of the proposed network.

by adding multiple nested edge detection (HED) [32] in the encoding stage and fusing the results. Although such methods can enhance the building boundary, the additional edge branches also increase the time consumption. Besides, postprocessing techniques are also widely used to optimize the building boundary. Liu et al. [20] integrated CNN with Markov random field (MRF) to achieve deterministic end-to-end computation. Shrestha and Vanneschi [33] added conditional random fields (CRFs) to the end of the CNN to sharpen the boundary and reduce the noise. In addition, some methods use auxiliary data like light detection and ranging (LiDAR) [34] and digital surface model (DSM) [7] to guide boundary optimization. Unfortunately, the data costs and requirements greatly limit their availability.

C. Corner Optimization of Building Extraction

In building extraction, most of the work that focuses on corner optimization belongs to buildings' automatic delineation or annotation. They take the coordinate sequence of the initial vertex or polygon as an input and use CNN [35], [36], long short-term memory (LSTM) [37], and recurrent neural networks (RNNs) [38], [39], for the next vertex or optimized polygon prediction. A segmentation process is first required to give them the initial vertices or polygons, followed by an additional network to learn the contours. Moreover, unlike the semantic segmentation task, these methods also require the annotation of the building's actual vertices.

III. Methodology

A. Overview

As illustrated in Fig. 2, our proposed method is a three-stage network. It comprises four components: the backbone network, the decoder with LDSC blocks, the DFCM, and the MSPRM. As shown in Algorithm 1, the pseudocode describes a three-stage building segmentation algorithm that progressively optimizes the building integrity, boundary, and corners.

In this article, a hybrid structure R50-ViT [40] consisting of ResNet50 and the vision transformer (ViT) is used as the default backbone. We use the backbone as an encoder to extract deep features. The designed network greatly boosts the reuse of features, and multiresolution feature maps are optimized at each stage. 1) In the first stage, the LDSC-decoder recovers detailed information from the feature maps output from the backbone by convolution and cascade upsampling and generates the initial segmentation. Low-level features from the backbone are connected to the decoder to retain high-frequency information. Notably, LDSC blocks are plugged into the decoder to improve the building's integrity. The multiresolution feature maps optimized by LDSC blocks in this stage are passed on to the next stage. 2) In the second stage, the DFCM learns the direction field to correct the multiresolution feature map and control the building's overall shape in the initial segmentation. The optimized segmentation and the corrected feature maps are fed into the next stage. 3) In the final stage, MSPRM selects boundary and corner points via multistrategy. Point features on multiresolution feature maps are searched to support reclassification to further optimize boundaries and redetermine corners. The output mask of MSPRM serves as the final building extraction result of IBCO-Net. In addition, the initial segmentation, optimized segmentation, direction field, and point classification results from the above process are used to compute losses to supervise the co-optimization of the integrity, boundaries, and corners.

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Algorithm 1 Training Process of IBCO-Net

Data: Image and ground truth (I, G)
Result: Trained model

while training **do**
  $D \leftarrow$ Backbone(I)
  Decoding Stage:
    \\ Obtain the list of feature maps
    $F \leftarrow$ LDSC-decoder(D)
  Direction Field Correction:
    Get the initial segmentation IS
    \\ Step1:Predict direction field from multi-directional feature maps and get direction truth R
    $F_{df} \leftarrow$ Conv_1 $\times$ 1(F_{d1}, F_{d2}, F_{d3}, F_{d4})
    $R \leftarrow$Distance transform(G)
    \\ Step2:Correct feature maps
    $F^0(x, y)$, $F^0_{df}(x, y) \leftarrow Grid(F, F_{df})$
  for $i = 1, \ldots, N$ **do**
    $F^i \leftarrow$ Correct($F^{i-1}(x, y)$, $F^{i-1}_{df}(x, y)$)
    $F^i \leftarrow$Grid_sample(IS, F^i)
    Get the optimized segmentation OS
  end for
  Multi-strategy Point Refinement:
    \\ Step1:Select boundary, corner, and normal point positions
    $P_P \leftarrow BP + CP + NP$
    \\ Step2:Get point features
    $F_P \leftarrow F^N(P_P)$
    \\ Step3:Reclassify points and get the final segmentation
    $OS(P_P) \leftarrow$MLP (F_P)
  Calculate Loss and Update Parameters
end while

B. Long-Range Dependent and Spatial-Continuous (LDSC) Decoder

To improve the building integrity and as a basis for boundary and corner optimization, we proposed the LDSC-decoder, as shown in Fig. 3.

In building extraction, the lack of building integrity can be caused by two aspects. One is the lack of long-range dependence on the building components, which often leads to the building failing to be recognized as a whole, especially for large buildings. The second is the lack of spatial continuity inside buildings, where conventional receptive fields can only detect limited local information, which lacks coherence in representing buildings with large internal color and texture differentiation.

Inspired by the above, we designed LDSC to simultaneously establish long-range dependence and enhanced spatial continuity. The proposed LDSC is shown in Fig. 3, which contains a long-range dependent (LD) convolution and a spatial continuity convolution path. Assuming that large buildings or clusters of dense buildings dominate the image, it is necessary to recognize the dominant category for large areas within the image. In the LD convolution, we build up extended receptive fields, whose dilation rate grows exponentially with the feature map resolution, and the information under various receptive fields is aggregated. This extended receptive field allows long-range pixels to be involved in the convolution operation, as visualized in Fig. 3. The calculation is as follows:

$$F_{\mathrm{LD}} = \Psi(F + Fd + F2d) \tag{1}$$

where $F \in \mathbb{R}^{C\times H\times W}$ refers to the input features and F_d and F_{2d} denote the features obtained after the atrous convolution with the dilation rate d and $2d$, respectively. Ψ stands for the linear rectification function. The specific dilation is given as follows:

$$Dr = \left(\frac{r}{r_0}\right)^2 * [1, 2], r \in \left\{r_0, 2r_0, \ldots, 2^N * r_0\right\} \tag{2}$$

where D_r denotes the dilation rate adopted by the LDSC corresponding to the feature map with resolution r. r_0 denotes the minimum resolution of the feature maps. $N \in \{1, 2, 3\}$ denotes the number of feature maps in the decoding stage. Based on the established formulas, the dilation rates of the LD convolution for the first three LDSC blocks in the decoder are set to [1, 2, 4, 8], and [16, 32], respectively.

Notably, the last LDSC needs to process feature maps with a resolution equal to the original image, which requires high-level details. Overly, large dilation rates can damage the results, so the dilation rate of the last LDSC with LD convolution is still set to [16, 32].

However, the receptive field achieved by LD convolution is sparse, leaving problems, such as holes in the building interior. Hence, spatial-continuous (SC) convolution is used to enhance the building's interior continuity. We set this convolution path to a fixed, sawtooth wave-like dilation rate combination according to the HDC [41] principle, i.e., [1, 2, 5] to fill the voids left by single atrous convolution while expanding the receptive field. The receptive field achieved by this convolution path conforms to a Gaussian distribution, as shown in Fig. 3.

Finally, the feature maps output from the two convolution paths is concatenated and fused by 1×1 convolution to obtain features with complementary LD and spatial continuity. The final receptive field is visualized in Fig. 3.

C. Direction Field Correction Module (DFCM)

For the overall optimization of the building boundary, we designed the DFCM to correct the building pixels. The workflow of DFCM is illustrated in Fig. 4, which can be summarized into two parts: direction field prediction and feature correction.

1) Direction Field Prediction: The direction field follows the definition used in medical image segmentation [42], due to the similar problem of boundary blurring and inhomogeneous probability distribution in building segmentation. The direction field of a foreground pixel p is defined as a unit vector $\overrightarrow{\mathrm{DF}}(p)$ with the nearest boundary pixel b directed to p. Meanwhile, the $\overrightarrow{\mathrm{DF}}$ of the background pixels is set to 0. Thus, the direction field of each pixel p in the image domain Ω is given by

$$\overrightarrow{\mathrm{DF}}(p) = \begin{cases} \dfrac{\overrightarrow{bp}}{|\overrightarrow{bp}|} & p \in \text{foreground} \\ (0, 0) & \text{otherwise.} \end{cases} \tag{3}$$

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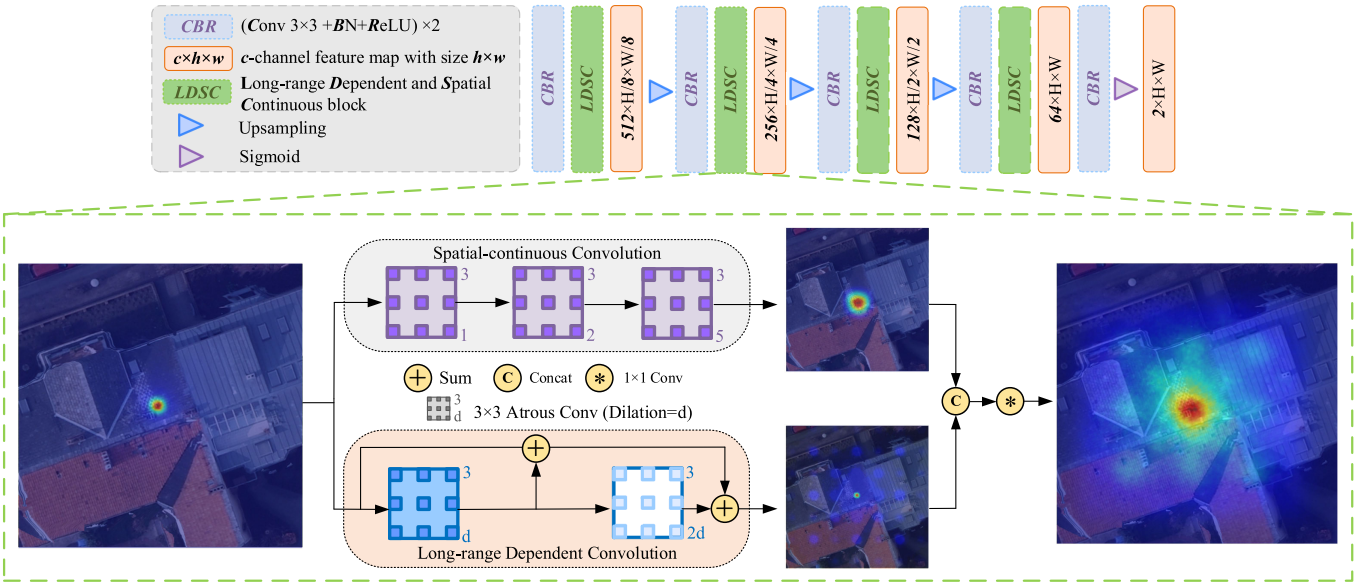


Fig. 3. Structure of LDSC-decoder. Conv + BatchNorm + ReLU (CBR) first reduces the channels of the feature map output from the backbone network and then expands the resolution by upsampling. This process is repeated four times to obtain the resolution-restored 64-channel initial segmentation and a two-channel mask. In particular, LDSC blocks are plugged in to improve the building's integrity. Each LDSC includes complementary LD convolutions and SC convolutions. The receptive fields of key steps are visualized and overlaid on the original image.

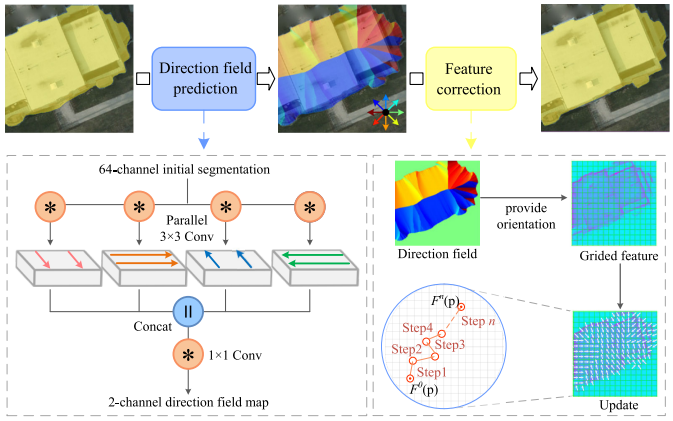


Fig. 4. DFCM uses a designed DFL to perform parallel convolution in four orientations on the initial segmentation to explore more spatial relationships and to infer the pixel-level direction field. The iterative algorithm uses the learned direction field $\overline{\mathrm{DF}}$ to assign an orientation $\overrightarrow{bp}$ to each pixel p. The features at the position pointed by $\overrightarrow{bp}$ are used to update $F(p)$. One update leads to one-step completion, and when N steps are completed, the boundaries of the initial segmentation are optimized.

Since each pixel's direction field is pointed to it by its nearest boundary point, implying the relative position of the building outline and the interior pixels, the building shape is encoded in this way. Notably, the direction field does not increase the manual labeling effort because direction field labels are transformed from mask labels by a distance transformation algorithm. Specifically, given a binary mask with a foreground pixel of 1 and a background pixel of 0, the distance transformation algorithm derives the minimum distance between each foreground pixel and its nearest background pixel.

The main distinction is that we designed a dedicated direction field learner (DFL) instead of using 1×1 convolution like [42] to learn the direction field roughly. Since visual cues in different directions can reveal building appearance to different degrees, the designed DFL explores more spatial relationships by detecting multidirectional subfeatures through parallel convolution in four directions. The detected multidirectional subfeatures facilitate 1×1 convolution to infer the pixel-level direction field.

2) Feature Correction: We use an iterative algorithm to correct and optimize the initial segmentation, as shown in Fig. 4. First, the initial segmentation and feature maps are gridded to obtain the position of each pixel. The learned direction field $\overrightarrow{\mathrm{DF}}$ provides an updating orientation $\overrightarrow{bp}$ for each pixel p. The features at the closest position to p in the orientation $\overrightarrow{bp}$ are used to update p. This process continues for N steps, with one update in one step. Step k is formulated as

$$F^k(p) = F^{(k-1)}\left(p_x + \mathrm{DF}(p_x),\, p_y + \mathrm{DF}\left(p_y\right)\right), k \in \{1, 2, \ldots, N\}. \tag{4}$$

When the N-step correction process is completed, the pixels near the boundaries are updated by the pixels that are N units away from them. It is worth noting that the N steps are not in a straight line because the current direction field changes with each correction step. Moreover, the DFCM in this article not only corrects the segmentation map but also corrects the multiresolution feature maps based on the learned direction field to make them obey a uniform regularity for the subsequent optimization stage.

D. Multistrategy Point Refinement Module (MSPRM)

To further refine the segmentation results, an MSPRM is designed to perform fine-grained optimization of the boundary and corner points.

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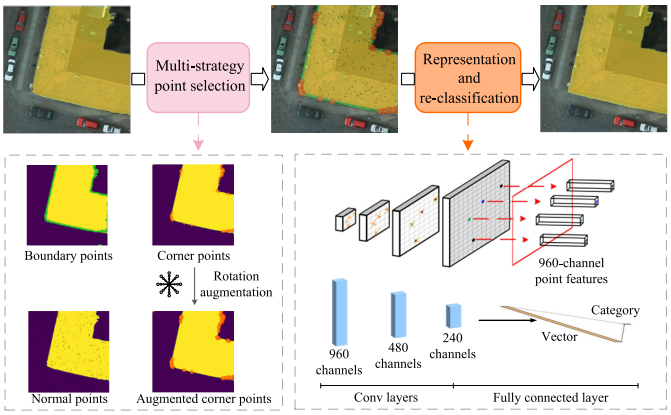


Fig. 5. Illustration of MSPRM. Selecting boundary and corner points via multiple strategies and representing their semantic-detail point features across feature maps. Using MLP to reclassify the selected points can further refine the boundary and relocate corners.

Three strategies for point selection are adopted in MSPRM. We first illustrate the three complementary point selection strategies depicted in the lower left corner of Fig. 5.

1) Boundary Points: Hard point sampling can help to precisely locate the fine-grained boundaries [43]. The building boundary generally has indistinguishable probability values between the foreground and background. Based on this premise, we select boundary points by designed rules without involving any edge enhancement and detection techniques. The models used for building segmentation eventually generate a two-channel feature map to describe the probability that each pixel belongs to the building or the background. We compute the difference between these two probabilities at each pixel and sort them, selecting the u pixels with the smallest difference as uncertain points during training. As shown in Fig. 5, these points are distributed around the boundary of the building. In prediction, the number of uncertain points is adaptive. The number of uncertainty points is determined based on the number of foreground pixels N_f in each prediction, i.e., $1/k$ of N_f (k is 10 by default).

2) Corner Points and Rotation Augmentation: The corners are also critical representations of the building morphology. The existing semantic segmentation methods hardly consider the optimization of corner point locations specifically. Unfortunately, a direct application of convolution also makes it difficult to determine which pixels belong to the corners of a building. We use oriented fast and rotated brief (ORB) to detect the building corners from the mask. This works by providing the network with the attention to know which pixels on the corners need to be considered. As an extension, we also treat the points in the eight directions around each detected point as augmented corner points, and these points are also regarded as potential corner points. Specifically, each detected corner point is used as an anchor point and rotated by $\theta = \{45°, 90°, 135°, 180°, 225°, 270°, 315°, 360°\}$ as its center to obtain another $8d$ key point within d pixels from it, and together with the original corner points, a total of $(8d + 1)$ corner points are obtained. This rotation augmentation can resist inaccurate corners in segmentation results and slight errors in detection and take more potential corners into account.

3) Normal Points: To increase the number of points available for training to maintain stability, we also select some random points on the segmentation results. These points are normally distributed to keep the ratio of foreground to background pixels close to the segmentation result. This strategy is only used for training and is not retained for prediction.

The workflow of MSPRM is illustrated in Fig. 5. First, MSPRM applies three strategies to generate points on a two-channel feature map Mfinal $\in \mathbb{R}^{2\times H\times W}$. In addition, the other four feature maps ($M1 \in \mathbb{R}^{512\times H/8\times W/8}$, $M2 \in \mathbb{R}^{256\times H/4\times W/4}$, $M3 \in \mathbb{R}^{128\times H/2\times W/2}$, and $M4 \in \mathbb{R}^{64\times H\times W}$) corrected by the DFCM are input. Among them, the lower the resolution, the more it tends to represent the global semantic information; the higher the resolution, the more it tends to represent the local detail information. The MSPRM searches for subfeatures from these feature maps at the corresponding locations through bilinear interpolation based on the selected points, and they are concatenated to obtain 960-channel pointwise features. These features contain semantic-to-detail information and can be considered 1-D vectors. They are then fed to a simple multilayer perceptron (MLP) classifier consisting of 1-D convolutional layers and a fully connected layer for reduction and classification, respectively. The category probabilities rederived by the classifier at the selected points are used to replace the original probabilities on the two-channel feature map.

So far, the segmentation result after point refinement is given.

E. Hybrid Loss Function

To supervise the network for integrity-boundary-corner optimization, we build a hybrid loss.

The hybrid loss comprises four sublosses supervising the initial segmentation of LDSC-decoder output (L_{IS}), the optimized segmentation of DFCM output (L_{OS}), the direction field predicted by DFCM (L_{DF}), and the points reclassified by the MSPRM (L_{points}). The total loss function is given as follows:

$$L_{\text{total}} = w1L_{\text{IS}} + w2L_{\text{OS}} + w3L_{\text{DF}} + w4L_{\text{points}}. \tag{5}$$

For initial segmentation loss L_{IS} and optimized segmentation loss L_{OS}, we choose dice loss [44] as the loss function due to its excellent performance on the binary classification task. The dice loss is given by

$$L_{\text{Dice}} = 1 - \frac{2 * \sum_i^N pi * gi}{\sum_i^N p_i^2 + \sum_i^N g_i^2} \tag{6}$$

where p_i is the predicted probability value of the ith pixel of the image, gi is the true value of the ith pixel of the image, and N is the total number of pixels of the image.

The morphology of buildings in the real world is regular and angular, with direction field characterizing this property well and can provide additional constraints. For direction field loss L_{DF}, we calculate the sum of Euclidean distance and the angular distance between the predicted direction field DF and

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TABLE I
OVERVIEW OF DATASETS IN COMPARATIVE EXPERIMENTS

Datasets	Source	Resolution	Tiles	Size	Covered area	Buildings
Massachusetts	Aerial images	1.00m	5935	256×256	~340km^2	~270000
Potsdam	Aerial images	0.05m	3800	512×512	~2.5km^2	~5800
Chinese cities	Satellite images	0.29m	7260	500×500	~152km^2	63886

the true direction field $\mathrm{DF}^{\sim}$. The formula is as follows:

$$L_{\mathrm{DF}} = \sum_{p\in\Omega} \left(||\mathrm{DF}(p) - \tilde{\mathrm{DF}}(p)||_2 + ||\cos^{-1}\langle \mathrm{DF}(p), \tilde{\mathrm{DF}}(p)\rangle||_2 \right). \tag{7}$$

For point reclassification, we introduce binary cross-entropy (BCE) loss [45], which is formulated as follows:

$$L_{\mathrm{points}} = -\sum_{i}^{N} pi * \log(g_i) + (1 - pi) * \log(1 - g_i) \tag{8}$$

where N denotes the total number of the selected points, while pi and gi denote the probabilities of the ith point's prediction and ground truth, respectively.

IV. EXPERIMENTS AND DISCUSSION

A. Datasets

To evaluate the proposed method's performance for building segmentation from various scenarios and multiresolution remote sensing images, we conducted experiments on the following three open datasets listed in Table I.

1) Massachusetts Building Dataset: We selected the Massachusetts dataset for our experiments to evaluate the proposed method's potential for massive building extraction from meter-level resolution data. This dataset was published in [46] and consists of 151 aerial images of the Boston area, each with a size of 1500 × 1500 and an area of 2.25 km^2. Thus, the entire dataset covers approximately 340 km^2. The dataset is randomly divided into a training set of 137 images, a test set of ten images, and a validation set of four images. We cropped these images into 5935 tiles of size 256 × 256 for training, validation, and testing.

2) ISPRS Potsdam Dataset: The ISPRS Potsdam dataset [47] was also included in the experiments to validate the proposed method for building fine segmentation capability. It contains 38 aerial images of size 6000 × 6000 with a resolution of 0.05 m. The original labels contain six land cover categories, and we retain only the building category and label all other categories as background to adapt to the building extraction task. We crop the images with R, G, and B bands into tiles of size 512 × 512 and filter out the tiles without buildings. Finally, 2477, 300, and 500 tiles are obtained for training, validation, and testing.

3) Dataset of Building Instances of Typical Cities in China: We selected the Chinese cities dataset [48] for our experiments to test the proposed method's ability to extract buildings from different regions from satellite images. This dataset contains 7260 images of size 500 × 500 from four cities, including Beijing, Shanghai, Shenzhen, and Wuhan. Among them, 5985 and 1275 are divided into training and test set, respectively. The original data source is free Google-supplied 19-level satellite images with a ground resolution of 0.29 m. It was originally used for instance segmentation. Since it also provides semantic segmentation labels, it is convenient to use this dataset for semantic segmentation tasks. During training, we randomly select 10% from the training set for validation.

B. Experimental Setup

1) Evaluation Metrics: In the evaluation section, we adopt the four most commonly used metrics for building extraction tasks, i.e., precision, recall, F1-score, and intersection over union (IoU), which are formulated as, respectively

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{9}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{10}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{11}$$

$$\text{F1-score} = \frac{2 * \mathrm{Precision} * \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{12}$$

where TP, FP, and FN denote the true positive, false positive, and false negative, respectively.

The above metrics are mainly influenced by the building's main body but are not sensitive to the boundary and corners. We introduce the boundary F-score (BdF) [49] and Hausdorff distance (HD) [50] to evaluate the boundary quality and corner accuracy. In particular, to exclude the interference of over-abundant background pixels and failed predictions, we exploit BdF and HD as object-level evaluation metrics. As shown in Fig. 6, we first obtain the contour of each building object by computing the connected domain and further obtain object boxes. If a building object can match a building in the label (IoU of the box ≥ 0.5), then BdF and HD are calculated from their contours.

BdF calculates the F1-score in the range of $1 - k$ pixels around the ground-truth boundary according to the following equation:

$$\mathrm{BdF} = \frac{1}{k}\sum_{T=1}^{k} \frac{2 * PT * RT}{RT + RT} \tag{13}$$

where $T = \{1, 2, \ldots, k\}$ is the threshold value around the ground-truth boundary, and k is the maximum threshold ($k = 5$ in this article). PT and RT are the precision and recall under the threshold T, respectively.

HD is used to measure the maximum–minimum distance between two sets of vertices, one from the ground truth and

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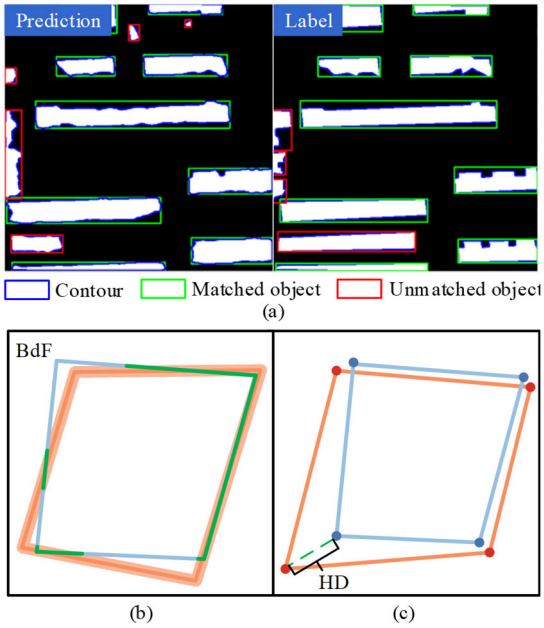


Fig. 6. Object-level boundary and corner evaluation metrics. (a) Excluding the interference of excessive background pixels and failed predictions by matching the building's object box with the corresponding building object on the label. (b) BdF for evaluating boundaries. (c) HD for evaluating corners.

the other from the contour of the predicted building. Given two sets of vertices from ground truth and prediction, i.e., $G = \{g1, g2, \ldots\}$, $P = \{p1, p2, \ldots\}$. HD is given by

$$\begin{aligned} \mathrm{HD}(P, G) &= \max[d(P, G), d(G, P)] \\ &= \max\left\{ \max_{p \in P}\left[\min_{g \in G} ||p - g||\right], \max_{g \in G}\left[\min_{p \in P} ||g - p||\right] \right\} \end{aligned} \tag{14}$$

where $d(P, G)$ and $d(G, P)$ are denoted as the forward and backward HDs between P and G, respectively.

2) Implementation Details: To fit the GPU memory, we cropped the images in the Massachusetts buildings dataset and Potsdam dataset using sliding windows of 256 × 256 and 512 × 512 size, respectively. For the dataset of building instances of typical cities in China, we used 500 × 500 images from the original dataset.

We train models on the Massachusetts buildings dataset, the Potsdam dataset, and the dataset of building instances of typical cities in China with the batch size set to 8, 4, and 4, respectively. We used the Adam optimizer in all experiments, and the initial learning rate was set to 0.001. The dice coefficients were verified every epoch, and if the dice coefficients did not increase for three consecutive epochs, then the current learning rate will be multiplied by 0.1. The models in this study are implemented based on PyTorch 1.9.0 with CUDA 11.4, and the GPU is an NVIDIA GeForce RTX 3090 24 GB. Each model used in the experiments was trained for 100 epochs. The direction field labels are generated online based on the mask labels during data loading in the training process.

C. Comparative Experiments

To evaluate the performance of the proposed method, we conducted comparative experiments with eight state-of-the-art methods on three open datasets from different regions and with different resolutions. These methods include four generic semantic segmentation models, including UNet [51], DeepLabv3+ [52], HRNet [53], and TransUNet [40]. Besides, four methods for building extraction have been proposed in recent years, including BRRNet [54], AGNet [55], DRNet [25], and STEB-UNet [56], which are also used for comparison.

1) Qualitative Comparison: Some representative examples of the visual comparison results are shown in Figs. 7 and 8, where the dashed lines indicate some noteworthy areas.

We can see that the two classical semantic segmentation models, UNet and DeepLabv3+ (columns 2 and 3 of Figs. 7 and 8), have a moderate performance, and the main drawback is that they tend to fail in segmenting some large buildings having roofs with color and texture differentiation. This is also a significant drawback of the CNN-based semantic segmentation models, i.e., it is difficult to capture the global information by the limited receptive field. The later proposed HRNet (column 4 of Figs. 7 and 8) has a better representation of building details such as boundaries due to the design of a repetitive information interaction strategy between high-low resolution feature maps to improve localization accuracy. However, this strategy may cause more erroneous segmentation on high-resolution images with a lot of interfering background information [e.g., the Chinese cities dataset shown in Figs. 7(e) and (f) and 8(e) and (f)], because it repeatedly introduces intrusive information into the feature maps. The recently proposed TransUNet introduces the ViT to globally encode the features and apply multiple self-attention layers. Due to the transformer's good perception of low-frequency information, TransUNet (column 5 of Figs. 7 and 8) usually enables a more complete segmentation of large buildings compared to the CNN-based semantic segmentation approach. BRRNet (column 6 of Figs. 7 and 8) considers the multiscale characteristics of buildings and specially designs the residual refinement structure to improve the building extraction ability, but its improvement on the building integrity is sometimes not obvious visually. AGNet (column 7 of Figs. 7 and 8) extracts buildings with better integrity and smoother boundaries by introducing a gated attention mechanism to control skip feature connections. It indicates that gated attention effectively suppresses the useless information in remote sensing images. However, for large buildings dominating the image, the limited receptive field of AGNet still prevents it from getting a complete segmentation result. DRNet (column 8 of Figs. 7 and 8) introduces a dense fusion strategy based on DeepLabv3+ but does not gain visual improvement in its building extraction performance. The latest proposed STEB-UNet (column 9 of Figs. 7 and 8) integrates the shifted-window (Swin) transformer into the U-shaped network and achieves better building integrity. This is because large-scale semantic information is more easily captured by self-attention in the transformer.

Although the above methods achieve building segmentation based on CNN or transformer, they do not sufficiently focus

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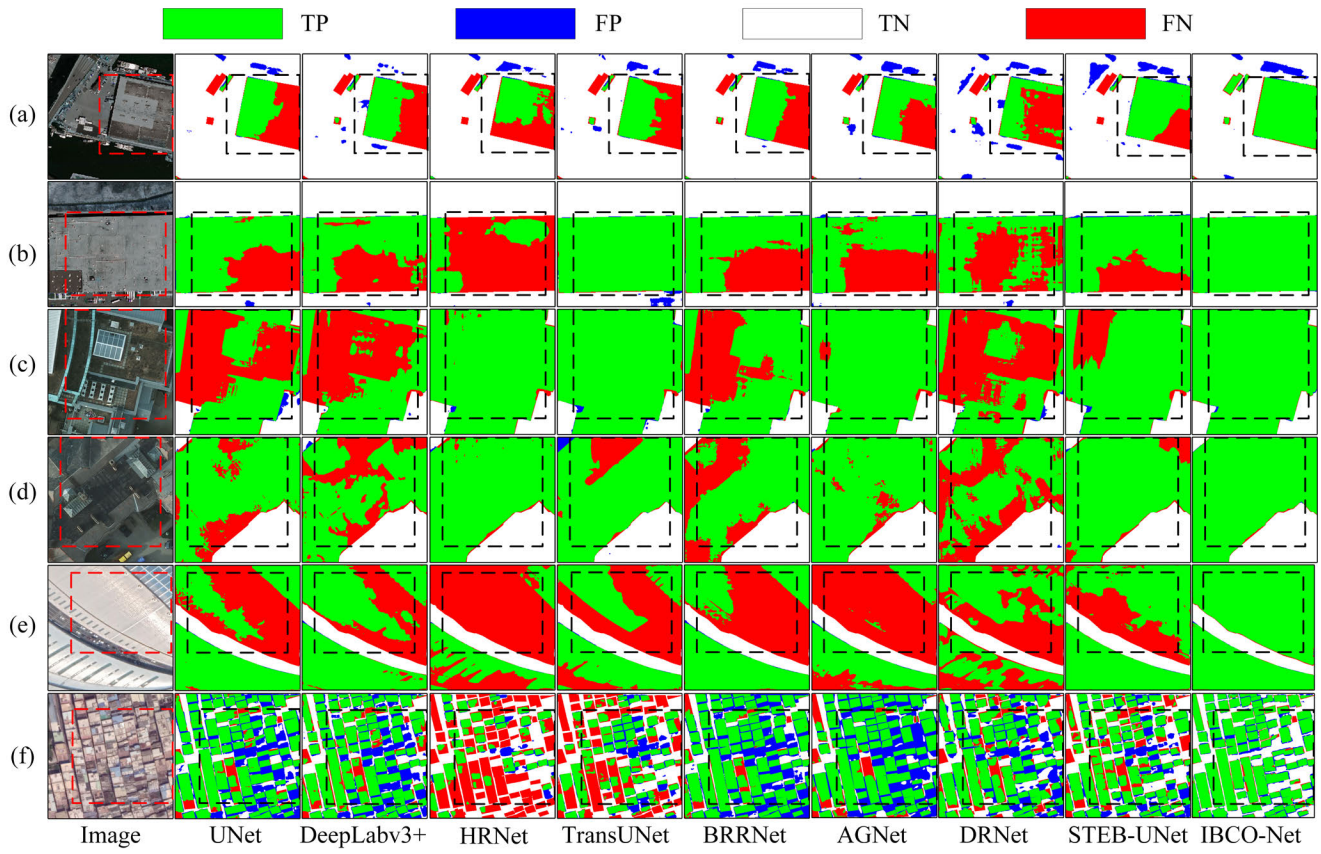


Fig. 7. Visual comparison of integrity. (a) and (b) Massachusetts dataset. (c) and (d) Potsdam dataset. (e) and (f) Chinese cities dataset.

on the characteristic of the building itself. In contrast, IBCO-Net (column 10 of Figs. 7 and 8) better preserves the building integrity, long-range dependence, and enhances spatial continuity, so that large buildings can be kept under a large enough receptive field and the internal continuity is improved. Second, IBCO-Net obtains more accurate and regular building boundaries. This is due to learning the direction field, iteratively correcting the features to control the building's overall shape, and further optimizing the boundaries through boundary point refinement. Third, IBCO-Net captures more accurate building corners. This results from the effective corner point selection and augmentation that allows the model to know the building corners of interest and reclassify them based on point features.

2) Quantitative Comparison: Tables II–IV present the quantitative evaluation results of the compared methods on the Massachusetts, Potsdam, and the Chinese cities dataset, respectively.

The experimental results show that the proposed IBCO-Net ranks first in the precision, recall, F1-score, and IoU on all three datasets, which indicates that IBCO-Net can predict the mask closest to the label. Especially, the proposed method shows superiority in F1 and IoU, due to the optimization of the building integrity. IBCO-Net achieves F1-scores of 85.47%, 95.86%, and 87.61% on the Massachusetts, Potsdam, and Chinese cities dataset, respectively. The highest F1-score shows that the proposed method well balances precision and recall and reduces misclassified and missing building pixels. The proposed method reaches IoUs of 74.73%, 92.25%, and 78.32% on the Massachusetts, Potsdam, and Chinese cities dataset, respectively. The highest IoU suggests that the proposed method obtains the buildings closest to the real world.

Moreover, the proposed method has significant advantages in two object-level metrics, BdF and HD. IBCO-Net improves BdF by 2.97%–8.53%, 1.8%–21.37%, and 1.68%–7.21% for the Massachusetts, Potsdam, and Chinese cities datasets, respectively. The improvement in BdF confirms that higher boundary quality can be obtained through overall shape control and fine-grained optimization. IBCO-Net reduces HD by 0.53–1.0, 1.93–31.65, and 1.52–3.55 for the Massachusetts, Potsdam, and Chinese cities dataset, respectively. The reduction in HD proves that we have improved the building corner accuracy by detecting, augmenting, and reclassifying corner points.

D. Robustness Analysis

The above three datasets are open datasets and most of them are three-band aerial images. To verify the robustness of the proposed method, we conducted a challenging experiment.

We built a dataset using four-band (blue, green, red, and near-infrared) GF-7 satellite images, and the multispectral images with a ground resolution of 2.6 m and the panchromatic images with a ground resolution of 0.8 m were fused. As shown in Fig. 9, the test and training sets have differences in the dominant building types, and such a dataset tests the robustness of the model more. We use the pretrained models on the Massachusetts dataset for this experiment and freeze the

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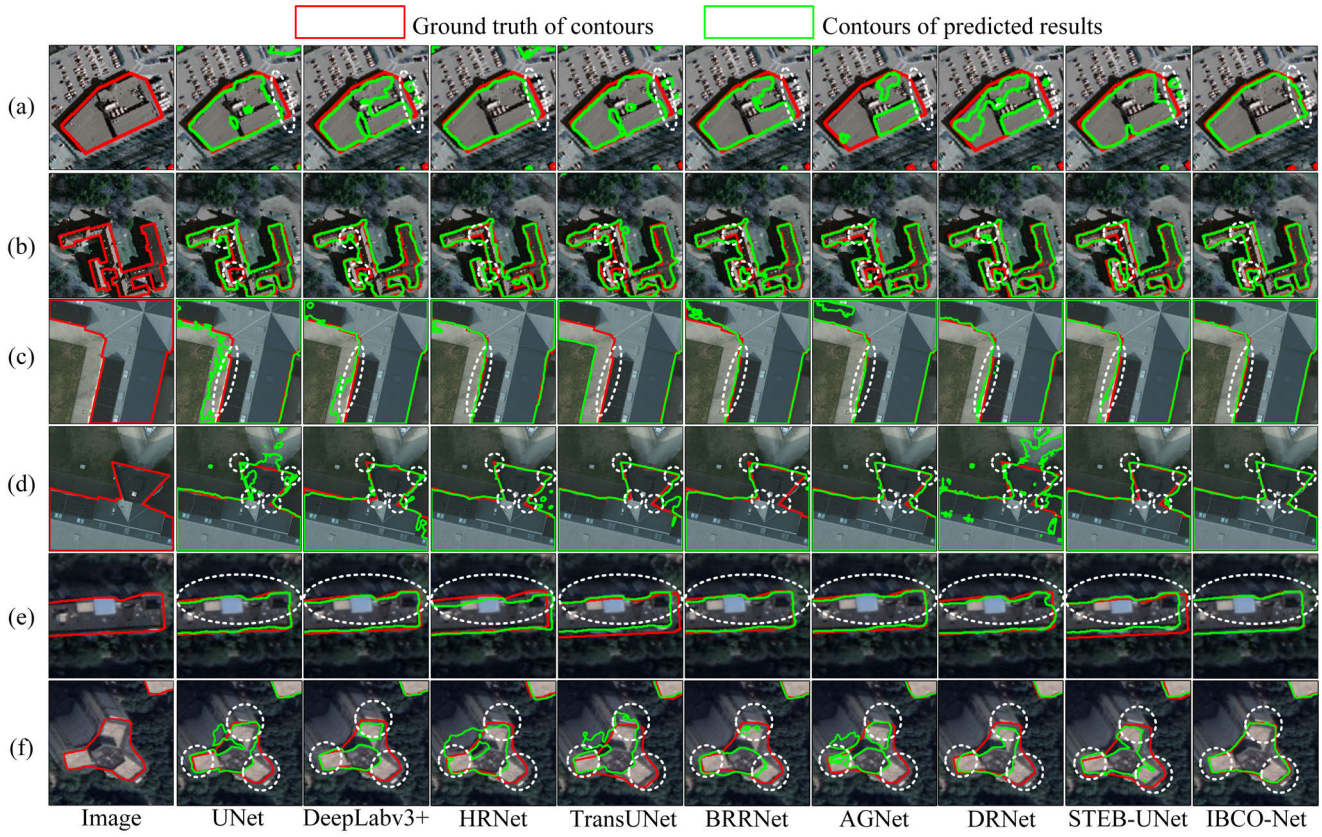


Fig. 8. Visual comparison of boundaries and corners. (a) and (b) Massachusetts dataset. (c) and (d) Potsdam dataset. (e) and (f) Chinese cities dataset.

TABLE II
QUANTITATIVE EVALUATION OF EXPERIMENTAL RESULTS ON THE MASSACHUSETTS BUILDING DATASET

Methods	Pixel-level metrics				Object-level metrics	
	P (%)↑	R (%)↑	F1 (%)↑	IoU (%)↑	BdF (%)↑	HD ↓
UNet	86.98	77.30	81.22	69.43	72.88	4.06
DeepLabv3+	85.00	73.72	78.09	65.32	67.91	4.47
HRNet	86.00	78.98	82.04	69.87	69.52	4.30
TransUNet	86.64	79.07	82.51	70.22	70.63	4.32
BRRNet	86.71	77.97	81.25	69.61	71.75	4.14
AGNet	84.42	76.54	79.57	67.01	70.83	4.22
DR-Net	83.83	74.65	78.24	64.89	67.32	4.53
STEB-UNet	86.22	78.88	81.52	70.13	70.75	4.25
IBCO-Net	**87.01**	**83.23**	**85.47**	**74.73**	**75.85**	**3.53**

Note. Bold font indicates the best performance of each column. ↑ denotes the metric that is better with higher values and ↓ denotes the metric that is better with lower values.

proposed modules and only train the backbone network for one epoch. This approach not only tests the models' robustness but also has important implications for urgent applications that lack training time. The quantitative evaluation of the experimental results is listed in Table V. We can see that even though the IBCO-Net's performance decreases compared to the above experiments, it is still the best in most of the metrics. This shows that the designed modules can work even when applying new datasets. In addition, we selected some representative examples, which are visualized in Fig. 10. As seen in the figure, IBCO-Net still has the advantage of obtaining the building's integrity, boundaries, and corners.

The above results show the robustness of IBCO-Net and its ability to adapt to new datasets.

E. Stability Analysis

In building extraction, a higher recall means that the building is extracted more completely, and a higher precision shows that the extracted building pixels include fewer misclassified pixels. Unfortunately, there is always a competing relationship between precision and recall. Hence, balancing precision and recall is an important challenge. Using the Massachusetts dataset as an example, we further analyzed the performance of the proposed method with the state-of-the-art

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TABLE III
QUANTITATIVE EVALUATION OF EXPERIMENTAL RESULTS ON THE POTSDAM DATASET

Methods	Pixel-level metrics				Object-level metrics	
	P (%)↑	R (%)↑	F1 (%)↑	IoU (%)↑	BdF (%)↑	HD ↓
UNet	83.67	92.00	85.55	77.17	60.40	74.21
DeepLabv3+	91.14	92.23	90.67	84.19	72.63	55.52
HRNet	94.23	92.49	92.80	87.37	78.16	44.67
TransUNet	93.49	94.30	93.41	88.37	77.89	52.16
BRRNet	91.69	94.19	92.38	86.47	71.33	54.31
AGNet	93.97	93.59	93.46	88.07	77.47	51.17
DR-Net	86.02	85.94	84.16	74.88	58.52	74.39
STEB-UNet	95.56	93.04	93.93	89.13	78.09	50.17
IBCO-Net	**97.06**	**94.91**	**95.86**	**92.25**	**79.89**	**42.74**

Note. Bold font indicates the best performance of each column. ↑ denotes the metric that is better with higher values and ↓ denotes the metric that is better with lower values.

TABLE IV
QUANTITATIVE EVALUATION OF EXPERIMENTAL RESULTS ON THE DATASET OF BUILDING INSTANCES OF TYPICAL CITIES IN CHINA

Methods	Pixel-level metrics				Object-level metrics	
	P (%)↑	R (%)↑	F1 (%)↑	IoU (%)↑	BdF (%)↑	HD ↓
UNet	85.22	87.23	85.50	75.34	75.83	23.99
DeepLabv3+	87.62	82.89	84.48	73.83	74.42	24.05
HRNet	87.65	81.36	83.63	72.55	74.39	24.27
TransUNet	87.46	82.46	84.21	73.41	74.43	24.11
BRRNet	83.72	87.38	84.76	74.25	75.01	24.14
AGNet	85.41	83.73	83.78	72.79	75.05	24.66
DR-Net	86.30	81.18	82.99	71.59	70.30	25.53
STEB-UNet	89.62	82.46	85.30	75.03	75.42	23.50
IBCO-Net	**89.01**	**86.94**	**87.61**	**78.32**	**77.51**	**21.98**

Note. Bold font indicates the best performance of each column. ↑ denotes the metric that is better with higher values and ↓ denotes the metric that is better with lower values.

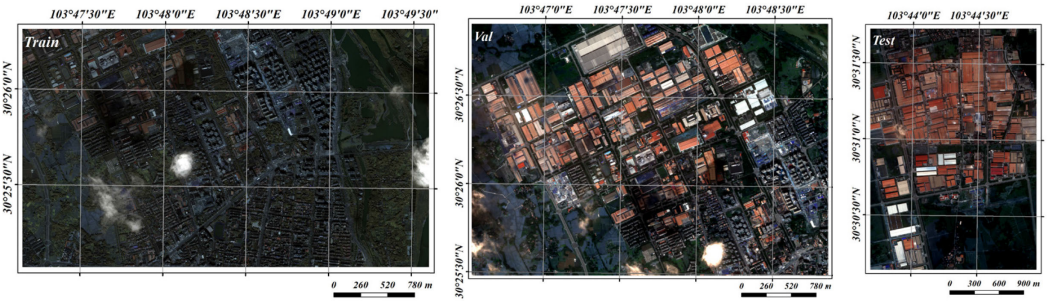


Fig. 9. Study area of self-labeled GF-7 satellite dataset. The training and validation sets are from neighboring regions and the test set is from another region.

method regarding balancing ability, as shown in Fig. 11. Two different curves are used to express the precision and recall, and the boxes between the curves indicate their differences. From this figure, it can be seen that the proposed IBCO-Net minimizes the difference between precision and recall to 4.75%. This result indicates that IBCO-Net can accommodate the completeness and correctness of the predicted building pixels to obtain a more comprehensive segmentation result.

To further verify the stability of the proposed method, we introduce Monte Carlo experiments, a random sampling technique, to judge whether the model is stable. We conducted 20 independent Monte Carlo experiments on the Massachusetts dataset. Specifically, we randomly discarded features by adding dropout after the first convolution layer of the LDSC-decoder and setting the probability to 0.5. This is a challenging experiment because discarding features at the first layer in the network affects the computation of subsequent layers. We counted six pixel-level and object-level metrics for each independent experiment in this extreme situation and calculated their average accuracy and standard deviation. As shown in Fig. 12, our method can still achieve high accuracy in independent Monte Carlo experiments. In particular, the standard deviations of each metric are small, with those of P, R, F1, IoU, and BdF below 0.15. Even for HD, a nonpercentage metric, the standard deviation is much smaller than the average value. The above results further demonstrate the stability of IBCO-Net.

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TABLE V
QUANTITATIVE EVALUATION OF THE EXPERIMENTAL RESULTS ON THE GF-7 DATASET

Methods	Pixel-level metrics				Object-level metrics	
	P(%)↑	R(%)↑	F1(%)↑	IoU(%)↑	BdF(%)↑	HD↓
UNet	82.22	73.72	75.36	60.47	64.24	11.86
DeepLabv3+	83.51	71.22	74.98	59.97	63.55	12.03
HRNet	68.26	75.54	76.61	60.79	62.25	12.77
TransUNet	80.94	73.96	76.91	62.48	65.91	11.73
BRRNet	80.85	74.87	75.06	60.07	63.71	12.62
AGNet	80.69	73.05	76.68	62.19	65.87	11.92
DR-Net	85.82	68.05	72.30	56.61	59.38	13.84
STEB-UNet	81.66	72.87	74.36	59.84	62.18	12.57
IBCO-Net	**85.15**	**76.39**	**76.92**	**64.55**	**66.95**	**9.97**

Note. Bold font indicates the best performance of each column. ↑ denotes the metric that is better with higher values and ↓ denotes the metric that is better with lower values.

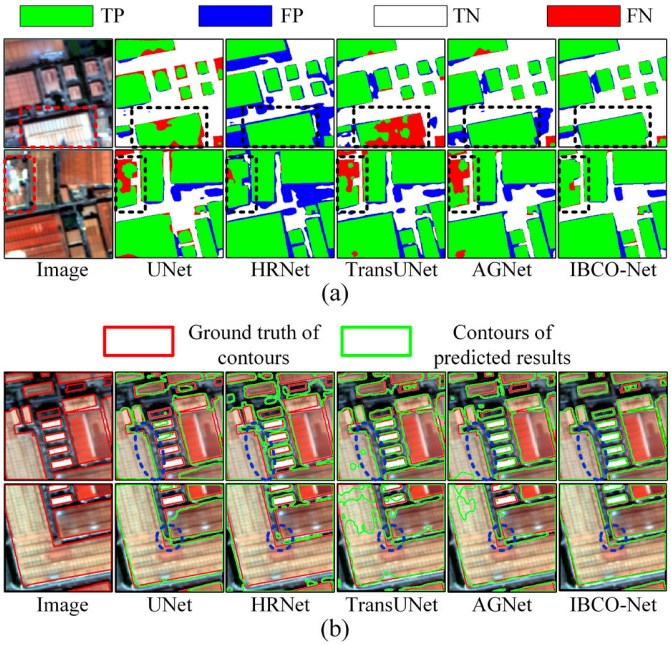


Fig. 10. Visual comparison on the GF-7 dataset. (a) Integrity. (b) Boundaries and corners.

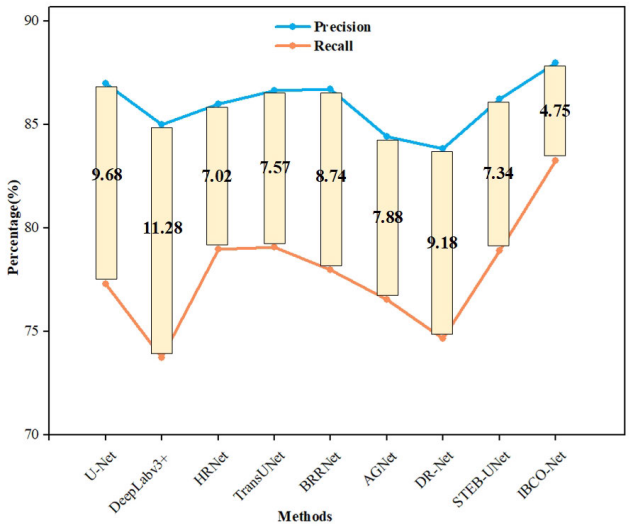


Fig. 11. Comparison of the differences between precision and recall.

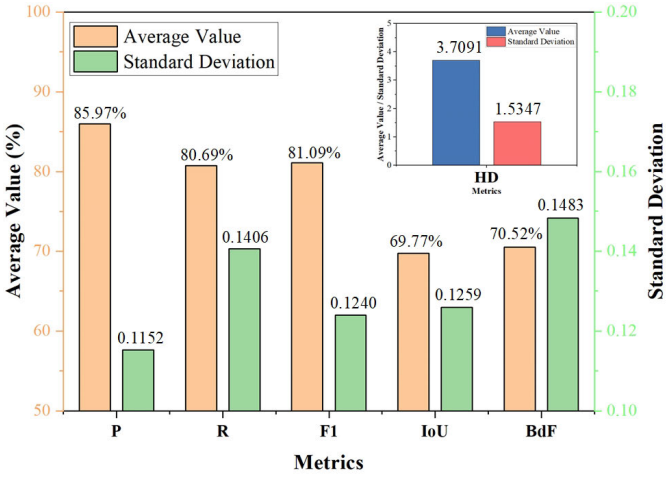


Fig. 12. Average values and standard deviations of Monte Carlo experiments.

F. Hybrid Loss Analysis

To explore the role of each subloss in the hybrid loss and to find the optimal weights, we conducted experiments using different combinations of weights in the Massachusetts dataset. The quantitative evaluation results are presented in Table VI.

(w_1, w_2, w_3, w_4) = (1, 1, 1, 1): First, we do not assign specific weights to sublosses but adopt equal weighting and use this approach as the baseline method. As shown in Table VI, the pixel-level metrics P, R, F1, and IoU can reach 84.89%, 82.61%, 84.03%, and 72.47%, respectively. The object-level metrics BdF and HD are 74.42% and 3.85.

(w_1, w_2, w_3, w_4) = (0.4, 0.4, 0.1, 0.1): Given that the direction field learning and point classification have equal magnitude, we set their weights w_3 and w_4 to 0.1 and assign equal weights to both the initial segmentation loss and the optimized segmentation loss, i.e., $w_1 = w_2 = 0.4$. R, F1, and IoU were boosted by 0.95%, 0.18%, and 0.26%, respectively. However, BdF and HD are impaired by 0.19% and 0.11%, respectively. The slight increase in pixel-level metrics and the slight loss in object-level metrics indicate that overly biased

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TABLE VI
QUANTITATIVE EVALUATION OF DIFFERENT HYBRID LOSS WEIGHTING

Methods	Pixel-level metrics				Object-level metrics	
	P (%)↑	R (%)↑	F1 (%)↑	IoU (%)↑	BdF (%)↑	HD ↓
(w_1,w_2,w_3,w_4) = (1,1,1,1)	84.89	82.61	84.03	72.47	74.42	3.85
(w_1,w_2,w_3,w_4) = (0.4, 0.4,0.1,0.1)	84.23	**83.56**	84.21	72.73	74.23	3.96
(w_1,w_2,w_3,w_4) = (0.3,0.5,0.1,0.1)	86.70	82.85	84.79	73.61	74.98	3.78
(w_1,w_2,w_3,w_4) = (0.2,0.4,0.2,0.2)	**87.01**	83.23	**85.47**	**74.73**	**75.85**	**3.53**

Note. w_1,w_2,w_3, and w_4 refer to the weights of initial segmentation, optimal segmentation, direction field, and point loss. Bold font indicates the best performance of each column. ↑ denotes the metric that is better with higher values and ↓ denotes the metric that is better with lower values.

building segmentation and ignoring boundary and corner optimization do not improve the network performance in general.

(w_1, w_2, w_3, w_4) = (0.3, 0.5, 0.1, 0.1): Based on (w_1, w_2, w_3, w_4) = (0.4, 0.4, 0.1, 0.1), we refine the weights of the segmentation loss and adjust the weight of the optimized segmentation loss to 0.5 as the main loss. This weighting obtains an overall improvement compared to the baseline method. The pixel-level metrics P, R, F1, and IoU are improved by 1.81%, 0.24%, 0.76%, and 1.14%, respectively. The object-level metrics BdF and HD increase and decrease by 0.56% and 0.07, respectively. This is because the emphasis on optimized segmentation reinforces the building extraction performance and provides a good basis for boundary and corner optimization.

(w_1, w_2, w_3, w_4) = (0.2, 0.4, 0.2, 0.2): Based on the above results, we first proceed to adjust the weight of the segmentation loss. The loss weight of the optimized segmentation is adjusted to twice the initial segmentation to emphasize the optimized segmentation, i.e., w_1 and w_2 are 0.2 and 0.4, respectively. Second, we further increase the weights of the direction field and point losses, so that they enjoy equal weights as the initial segmentation, i.e., $w_1 = w_3 = w_4$. In this case, all six pixel-level and object-level metrics are improved comprehensively compared to the baseline method. R, F1, IoU, BdF, and HD all reached the best. This is due to that further optimization of the building segmentation provides a better basis for boundary and corner optimization, which is also given more attention.

G. Ablation Study

To verify the influences of the various components in the proposed network, we conducted the ablation experiment on the Massachusetts buildings dataset.

IBCO-Net consists of four components: the backbone network, LDSC-decoder, DFCM, and MSPRM. We use the encoder–decoder structure consisting of the backbone network and the decoder as the baseline and put the proposed three modules on it, respectively, to experiment. The results of the ablation experiments are visualized in Fig. 13. The quantitative evaluation results of the ablation experiments are given in Table VII to further illustrate each module's role. Next, we analyze different combinations of modules based on the visualization and quantitative evaluation results.

1) Analysis of LDSC Blocks: As seen in Fig. 13(c), the segmentation results obtained by the baseline mainly have three problems: lack of building integrity, irregular boundaries, and inaccurate corners. After LDSC blocks are added, the building integrity is significantly improved owing to LDSC blocks' ability to have a larger and more continuous view. In quantitative evaluation, adding LDSC blocks has improved most metrics compared to the baseline. In particular, the R-value has improved significantly by 3.79%, the highest increase among all metrics. The significant increase in recall indicates that it greatly improves the building's integrity, which is consistent with the visualization results, and the F1 and IoU values increased by 1.44% and 1.73%, respectively.

2) Analysis of DFCM: From the visualization results shown in Fig. 13(d), when the DFCM is combined with the baseline, the irregular boundaries caused by the interference of shadows and roads are mitigated. However, the building integrity in this case still needs to be improved. Its effect is also indicated in the quantitative evaluation results. Due to the overall control of the DFCM on the relationship between building pixels, the DFCMs influence on the segmentation result is relatively balanced. This is reflected by increased P and R values of 0.78% and 0.75%, respectively. The joint improvement in precision and recall increased F1 and IoU values by 1.12% and 1.10%, respectively. The boundary quality is significantly improved due to DFCMs correction of the building pixels, which is reflected in the increase and decrease of BdF and HD by 1.51% and 0.15, respectively.

3) Analysis of MSPRM: After the MSPRM is added to the baseline, the boundary quality is also improved, although the effect is not as visually apparent as with the DFCM in Fig. 13. This is because it works based on every single point selected, allowing for fine-grained boundary optimization. In the corner accuracy, the role of MSPRM makes a significant improvement. In addition, since it can reclassify some false negative pixels, it is helpful in filling the holes in buildings. The quantitative evaluation results also state the above. When MSPRM is combined with the baseline, more pixels are classified to the true class. This reclassification results in a joint improvement of the P and R values by 0.61%

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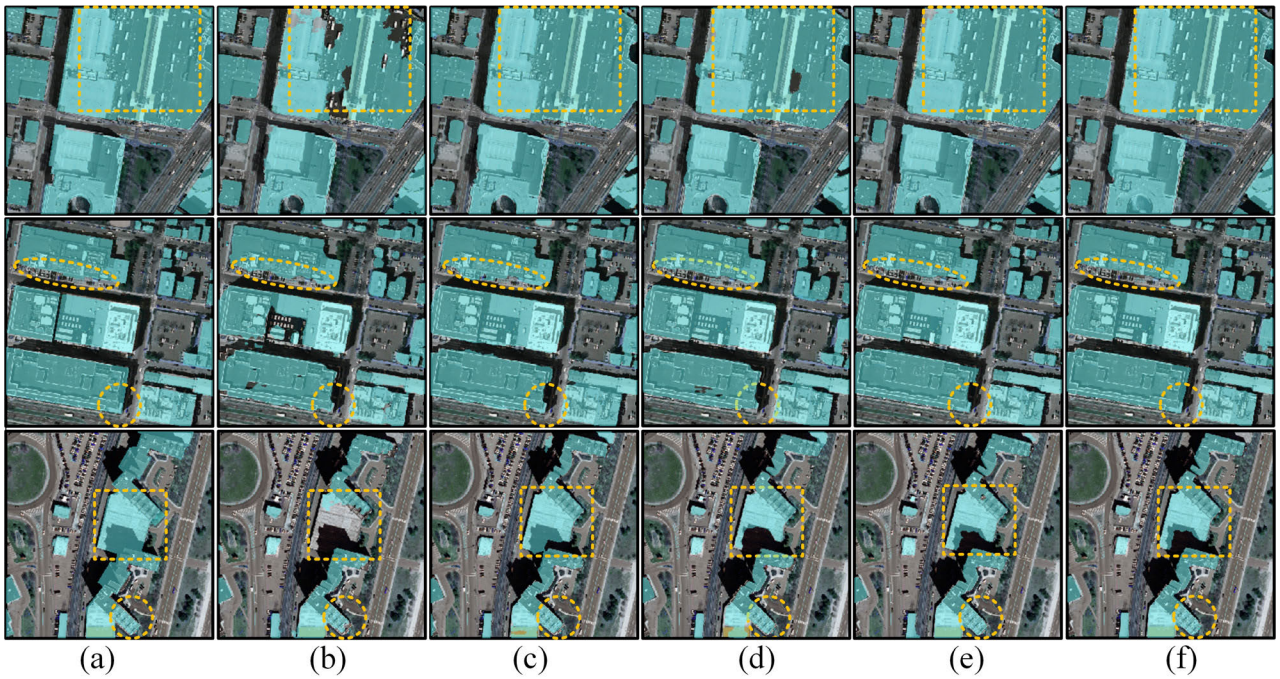


Fig. 13. Visualization of ablation experiment results. (a)–(f) Labeled image, backbone + the decoder without LDSC blocks (baseline), baseline + LDSC blocks, baseline + DFCM, baseline + MSPRM, and the proposed IBCO-Net, respectively.

TABLE VII
QUANTITATIVE EVALUATION OF ABLATION EXPERIMENTS

Methods	Pixel-level metrics				Object-level metrics	
	P (%)↑	R (%)↑	F1 (%)↑	IoU (%)↑	BdF (%)↑	HD ↓
Backbone + decoder (baseline)	85.61	79.32	82.35	70.43	70.82	4.30
+LDSCs	86.10 (+0.49)	83.11 (+3.79)	83.79 (+1.44)	72.16 (+1.73)	72.12 (+1.30)	4.12 (-0.18)
+DFCM	86.39 (+0.78)	80.07 (+0.75)	83.47 (+1.12)	71.53 (+1.10)	72.33 (+1.51)	4.15 (-0.15)
+MSPRM	86.22 (+0.61)	80.13 (+0.81)	83.45 (+1.10)	71.49 (+1.06)	72.25 (+1.43)	4.09 (-0.21)
IBCO-Net	**87.01** (+1.40)	**83.23** (+3.91)	**85.47** (+3.12)	**74.73** (+4.30)	**75.85** (+5.03)	**3.53** (-0.77)

Note. Bold font indicates the best performance of each column. ↑ denotes the metric that is better with higher values and ↓ denotes the metric that is better with lower values.

and 0.81%, respectively. Furthermore, the F1 and IoU values increased by 1.06% and 1.43%, respectively, indicating that MSPRM enables the model to obtain more comprehensive segmentation results. In the object-level metrics, the values of BdF and HD improve and decrease by 1.43% and 0.21%, respectively.

4) Analysis of the Proposed IBCO-Net: Finally, the model's performance is boosted best when we apply all the modules to the baseline to build the proposed network. The visualization results show that the building integrity, boundary quality, and corner accuracy have been improved. From the quantitative evaluation results, the P-value and R-value increase by 1.40% and 3.91%, respectively, compared to the baseline. The joint increase in P and R values suggests that our network has greatly improved by 3.12% and 4.30%, respectively, which indicates that IBCO-Net achieved excellent and comprehensive performance. In boundary metrics, the values of BdF and HD are increased and reduced by 5.03% and 0.77, respectively, more than the sum of the changes for each module. This result demonstrates their synergistic effect, allowing the proposed network to give the best boundaries and corners. Specifically, the optimization of the main body establishes the basis for the overall shape control, which with turn facilitates boundary and corner points' selection and reclassification.

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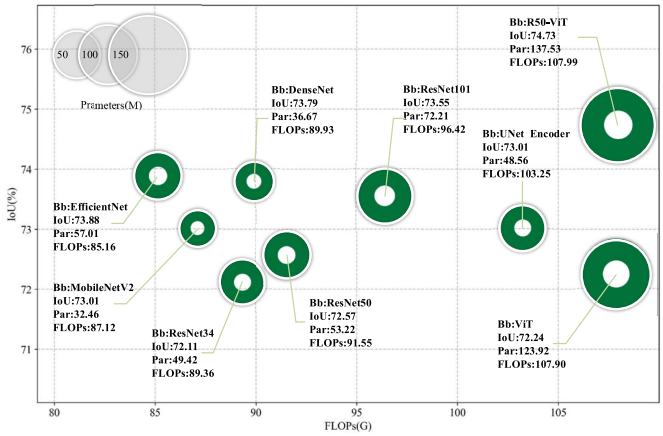


Fig. 14. Performance of the proposed IBCO-Net using nine existing CNN and transformer-based backbones. Bb, IoU, Par, and FLOPs mean backbone, intersection over union, parameters, and floating point operations, respectively.

H. Generality Analysis

To test the generality of the proposed method, we use nine state-of-the-art backbone networks to construct IBCO-Net. The backbone networks used in the experiments include UNet's encoder [51], ResNet34-101 [57], MobileNetV2 [58], EfficientNet [59], DenseNet [60], ViT [61], and R50-ViT [40].

The performance on the Massachusetts dataset using different backbone networks is shown in Fig. 14. We can see that the proposed method has stable and reliable performance when applying the existing CNN and transformer-based models as backbone networks. Especially, the IBCO-Net's performance is maximized when a hybrid network composed of ResNet50 and a ViT is used as the backbone. This is because the CNNs sensitivity to high-frequency information and the transformer's awareness of low-frequency information are well combined to provide more comprehensive features for IBCO-Net. In addition, when using lightweight networks, such as EfficientNet and DenseNet as the backbone, it can obtain relatively well. This saves computational costs and storage capacity, making it a more cost-effective choice.

The nine methods we experimented with provide different cost-focused and effect-focused options for the building extraction task, demonstrating the generality of IBCO-Net.

V. Discussion

A. Additional Experiments in Dense Scenarios

To test the proposed IBCO-Net's performance in large-scale dense scenarios, additional experiments are conducted on the aerial image subset of the WHU building dataset [62]. It covers an area of 450 km^2 with approximately 22 000 buildings. The RGB aerial image has a spatial resolution of 0.3 m. RGB aerial images with a spatial resolution of 0.3 m were cropped into 8189 tiles of 512 × 512 pixels. The dataset was then divided into a training set (4736 tiles and 130 500 buildings), a validation set (1036 tiles and 14 500 buildings), and a test set (2416 tiles and 42 000 buildings).

B. Experimental Setup

Since the dataset has been predivided into training, validation, and test sets and has provided 512 × 512 tiles, we followed this dataset setup. To accommodate GPU memory, we set the batch size to 4. We used the Adam optimizer in experiments, and the initial learning rate was set to 0.005. Dice coefficients were validated in each cycle, and if the dice coefficients did not increase in five consecutive cycles, then the current learning rate was multiplied by 0.2. The model for this experiment was implemented based on PyTorch 1.9.0 with CUDA 11.4, and the GPU was an NVIDIA GeForce RTX 3090 with 24 GB of memory. Since the size of the dataset is large enough, we randomly selected 3000 images from the training set for training each cycle to shorten the time per cycle and increase the number of training rounds. Each model used in the experiments was trained for 200 epochs.

C. Results and Evaluation

The visualization of the experimental results is shown in Fig. 15. Due to the enhanced LD and spatial continuity, some large buildings where other features appear on the roof can still be perceived more completely by the proposed IBCO-Net, as shown in Fig. 15(a). The proposed IBCO-Net can better distinguish some buildings too close to each other in a dense scene. Because the direction field can drastically change once they are stuck together, the additional supervision of the direction field can help avoid this error, as shown in Fig. 15(b). Furthermore, the visualization in Fig. 15(c) and (d) shows that the use of skip connections between the encoder and decoder, such as UNet, BRRNet, and TransUNet, allows for a more complete extraction of dense small buildings since the high-frequency information in the low-level features can better reveal the presence of dense and small buildings. Since the proposed IBCO-Net preserves the skip connection, it has comparable advantages with UNet, BRRNet, and TransUNet in extracting dense small buildings. In summary, due to its superiority in extracting large buildings and good performance on small buildings, the proposed IBCO-Net can still hold the highest recall in dense scenes, reaching 95.75% in Table VIII.

Such a leading recall drove the F1 and IoU improvements to 98.19% and 90.86%, respectively, ahead of comparison methods. In addition, the proposed IBCO-Net achieves the best BdF and HD of 83.51% and 6.13, respectively, as it controls the building's overall shape and refines the boundary and corner points. This advantage has been discussed on other datasets.

The above experimental results demonstrate the proposed IBCO-Net's effectiveness for integrity-boundary-corner optimization in large-scale dense scenarios.

D. LDSC Visualization

To reveal the optimization effect of LDSC on building integrity, Fig. 16 shows the visualization of the last LDSC block in the decoder trained on the WHU dataset.

We embed LDSC blocks to the decoder to improve the building's integrity. Before features are fed into the LDSC

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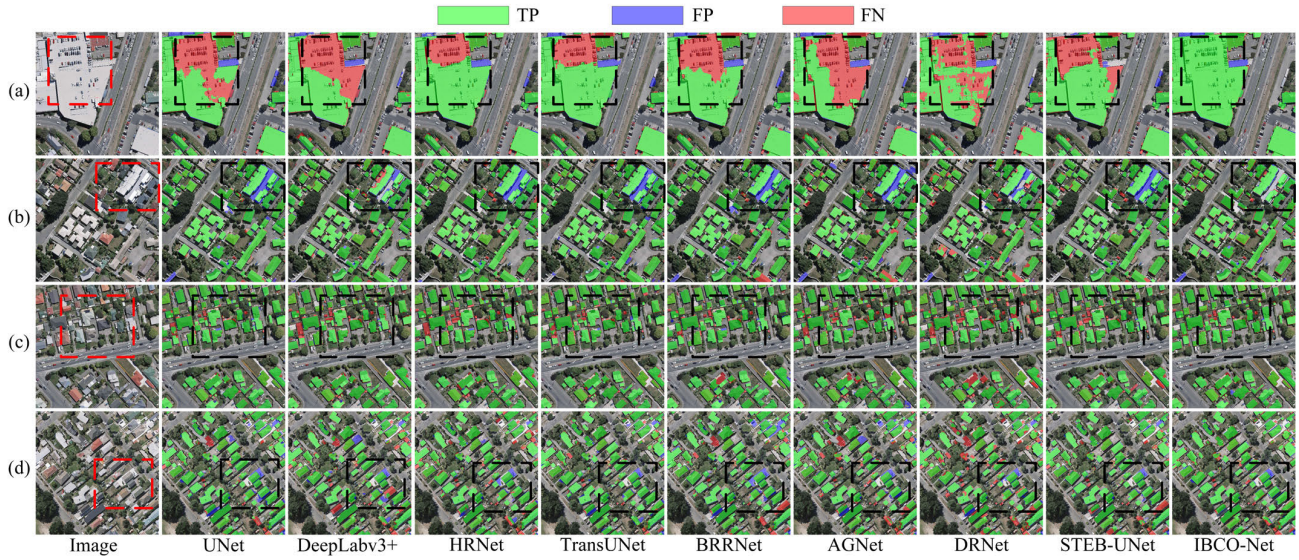


Fig. 15. Visualization of additional experiment results on the WHU dataset. (a) Large buildings. (b)–(d) Dense small buildings.

TABLE VIII
QUANTITATIVE EVALUATION OF THE EXPERIMENTAL RESULTS ON THE WHU DATASET

Methods	Pixel-level metrics				Object-level metrics	
	P(%)↑	R(%)↑	F1(%)↑	IoU(%)↑	BdF(%)↑	HD↓
UNet	90.86	94.28	92.33	86.07	80.85	7.18
DeepLabv3+	93.93	90.40	91.89	85.35	80.27	7.48
HRNet	91.98	92.31	91.89	85.36	81.93	7.36
TransUNet	93.22	92.16	92.48	86.35	81.91	7.08
BRRNet	91.10	93.17	91.88	85.32	80.67	7.33
AGNet	91.26	89.15	89.89	82.01	78.48	8.05
DR-Net	92.52	85.43	88.48	79.75	76.22	8.53
STEB-UNet	93.93	91.15	92.30	86.03	80.52	7.41
IBCO-Net	**94.68**	**95.75**	**95.19**	**90.86**	**83.51**	**6.13**

Note. Bold font indicates the best performance of each column. ↑ denotes the metric that is better with higher values and ↓ denotes the metric that is better with lower values.

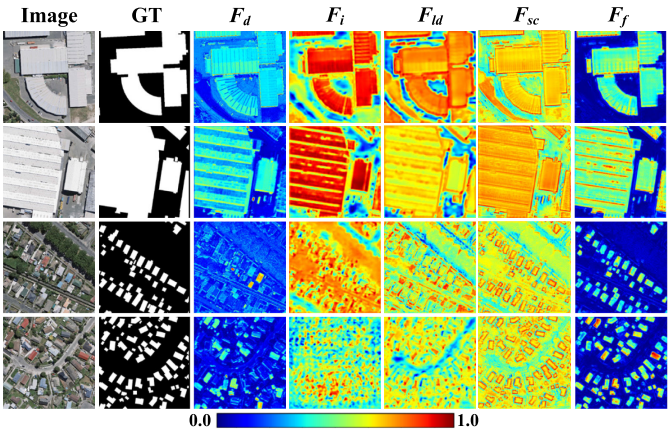


Fig. 16. Visualization of the LDSC trained on the WHU dataset. Taking the last LDSC in the decoder as an example, F_d, F_i, F_{sc}, F_{ld}, and F_f denote the detailed features of the skip connection to the decoder, input features, features of the SC convolutional path output, features of the LD convolutional path output, and fused features of the LDSC output, respectively.

block, the same-size feature maps in the encoder are skip-connected to the input for detail preservation and small object perception. From Fig. 16, the shallow feature F_d from the encoder can extract the high-frequency information representing the boundary and texture, which helps distinguish the dense buildings. Moreover, since F_d has a minor receptive field, some small buildings are more salient in the feature map. The last LDSC block's input feature F_i maintains good building integrity in the decoding stage due to the optimization by the first three LDSC blocks. Each LDSC block contains a LD convolution path and an SC convolution path. The LD path is devoted to establish the long-range dependence between regions in the image, and notably, the features of the original receptive field are added to this path's output to mitigate the omission of small objects by overly large receptive fields. The distinctive F_{ld} suggests dominant categories of large areas within the image, giving it a more complete perception of large-scale buildings. Besides, F_{ld} can distinguish extensive backgrounds, such as continuous roads near the foreground, which also requires a large receptive field. In addition, the SC path builds successive dilated convolutions, and the latter convolutional layer can cover the dots left by the previous convolutional layer. F_{sc} has enhanced spatial continuity in building interiors at various scales, showing homogeneously distributed eigenvalues complementary to F_{ld}.

As a result, the LDSC output F_f fuses the shallow feature F_d, LD F_{ld}, and SC F_{sc}, which incorporates their advantages.

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VI. Conclusion

In this study, we propose a general multistage network (IBCO-Net) for building fine segmentation from remote sensing images. It mainly addresses three problems: insufficient building integrity, irregular boundaries, and inaccurate corners. Comparative experimental results on three open datasets with different resolutions show that the proposed method achieves improvements in all the above three aspects. Experiments on our own annotated GF-7 satellite dataset have confirmed the good robustness of the proposed method and its adaptability to new datasets. The stability of the proposed method was tested by comparing the differences in precision and recall and conducting Monte Carlo experiments. The ablation experimental results show that LDSC blocks, DFCM, and MSPRM are effective in improving building integrity, enhancing boundary quality, and fine-grained optimization of boundary and corner, respectively. The cooperation of our proposed modules shows good compatibility and achieves the best building segmentation results. The cooperation performance with various existing backbone networks also verifies the proposed method's generality. Besides, additional experiments on the WHU aerial image dataset validate the proposed method in large-scale dense scenarios.

Nevertheless, this method still has a shortcoming. Since semantic segmentation usually gives segmentation results based on the whole image, leading to interference between individual building objects. This may impose a limitation on our proposed modules to play their boundary and corner optimization roles. In the future, we will consider developing our method into instance segmentation for more effective fine segmentation.

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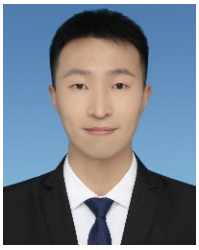
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